Lecture 5: Continuous Random Variables and Limit Theorems

Math for Data Science · 2026

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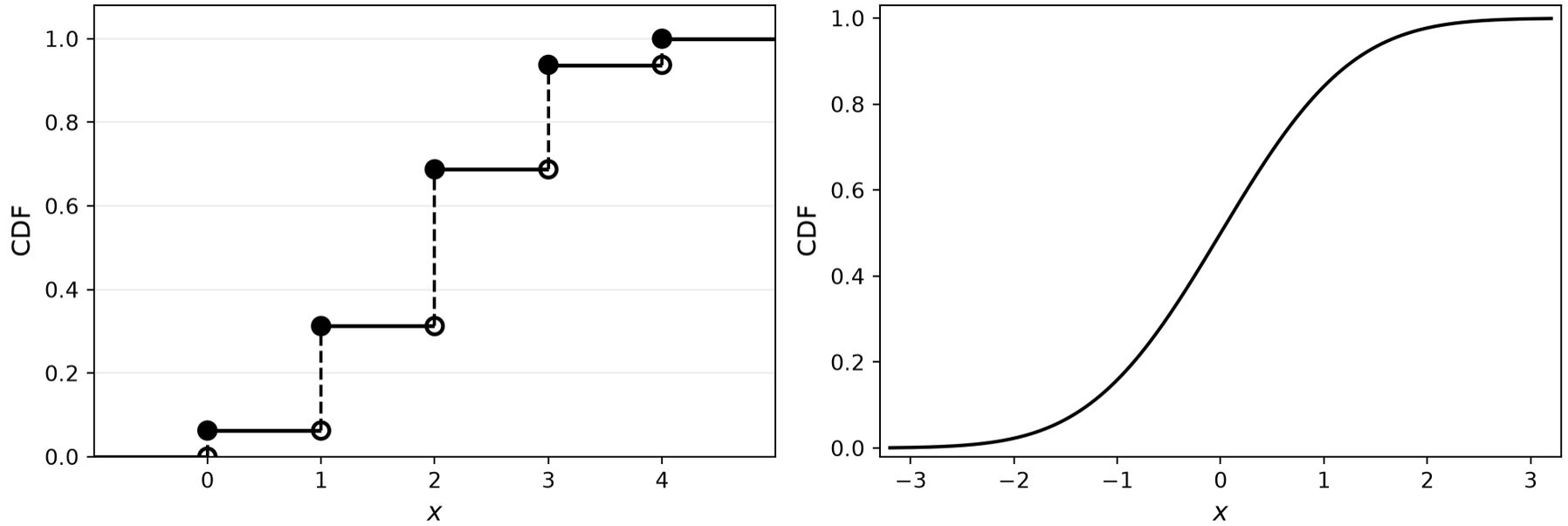
Agenda

- Part I: Continuous Random Variables
- Part II: Three Continuous Distributions
- Part III: Continuous Joint Distributions
- Part IV: Covariance and Correlation

Part I: Continuous Random Variables

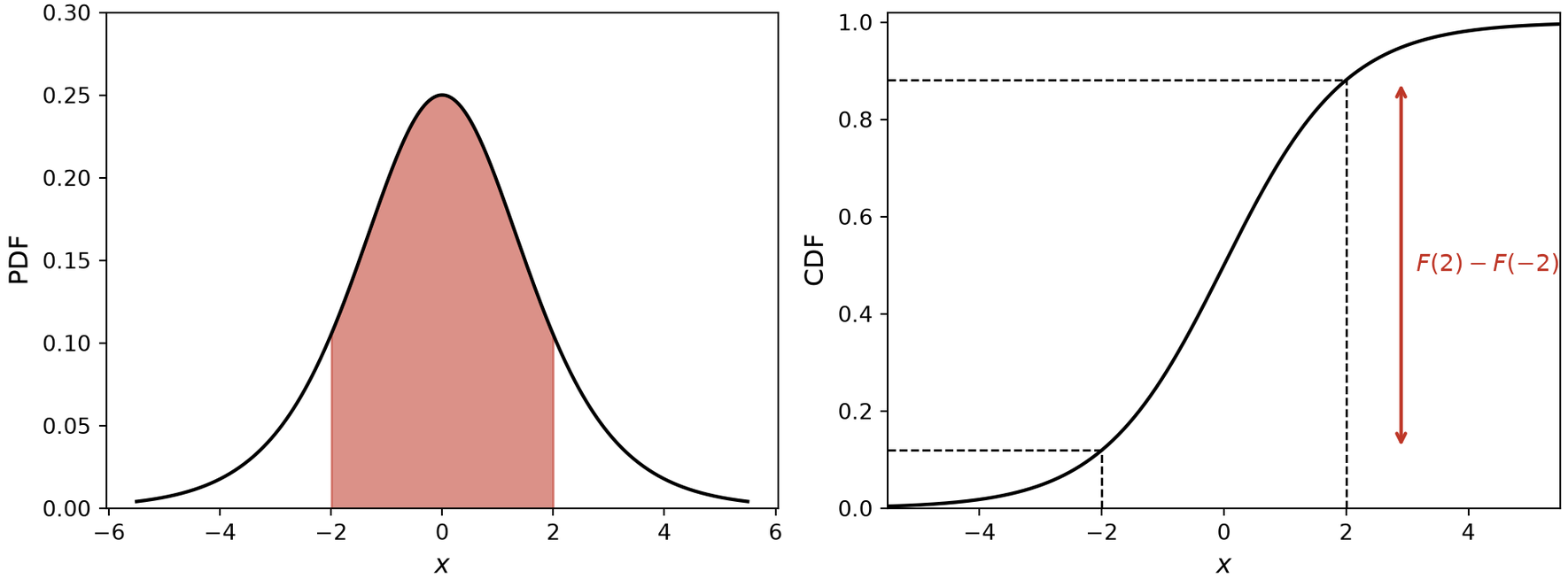
Continuous Random Variables

A random variable has a continuous distribution if its CDF is differentiable.



Left: CDF of a discrete r.v.; right: CDF of a continuous r.v.

The CDF of X is the Integral of the PDF of X!



The PDF of X must integrate to 1 over the support of X. Why?

The CDF of X is the Integral of the PDF of X!

For a continuous r.v. X with CDF F, the probability density function (PDF) of X is the derivative f of the CDF, given by:

$$f(x) = F'(x)$$

The support of X is the set of all x where $f(x) > 0$.

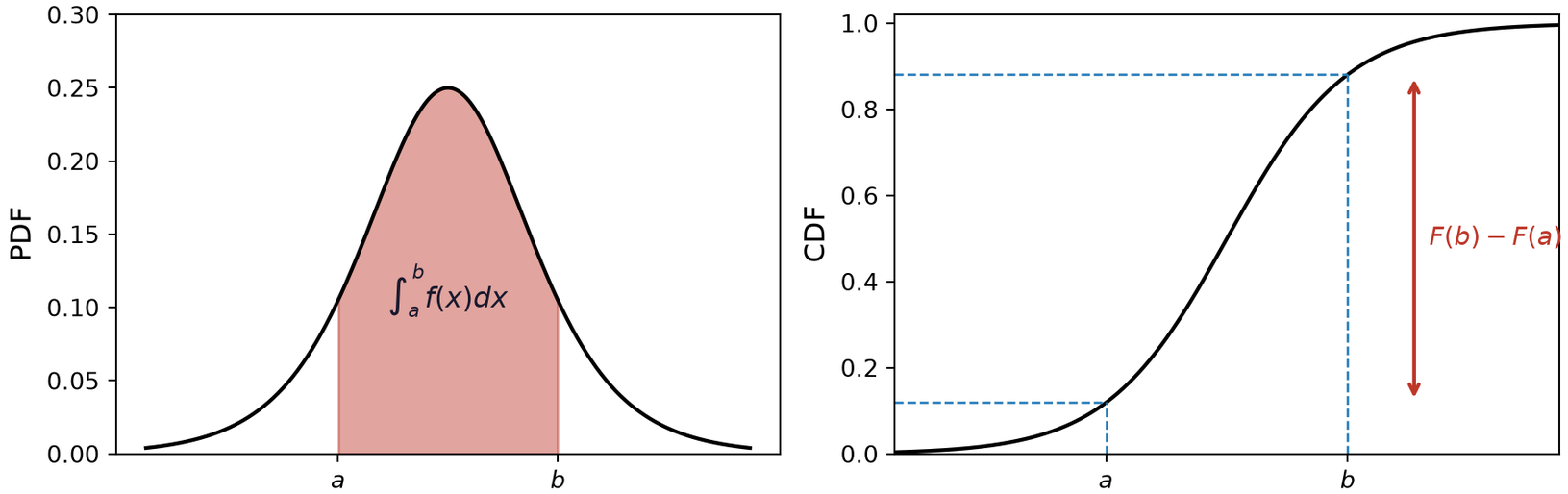
An important way in which continuous r.v.s differ from discrete r.v.s is that for continuous r.v.s, $P(X = x) = 0$ for all x.

- Why? Continuous r.v.s can take on infinite values.
- But that does not mean $f(x) = 0$! We simply do not interpret the PDF of X as $P(X = x)$.
- However, the CDF remains interpretable as $P(X \leq x)$.

Probability of a Continuous Random Variable

Instead, we are interested in the probability of X falling in some interval (a, b) - or $[a, b]$, or $(a, b]$, or $[a, b)$:

$$P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)\,dx$$



Valid PDFs

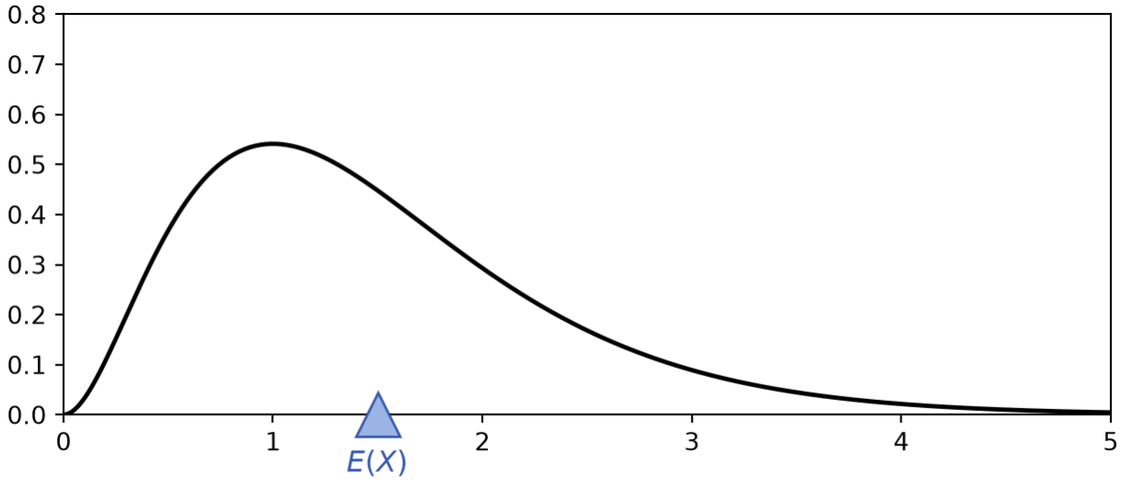
A valid PDF must satisfy two conditions:

1. **Nonnegative:** $f(x) \geq 0$
2. **Integrates to 1:** $\int_{-\infty}^{\infty} f(x)\,dx = 1$

Expectation of a Continuous Random Variable

The mean or expected value of a continuous random variable is defined as:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$



Part II: Three Continuous Distributions

Meet the Continuous Uniform Distribution

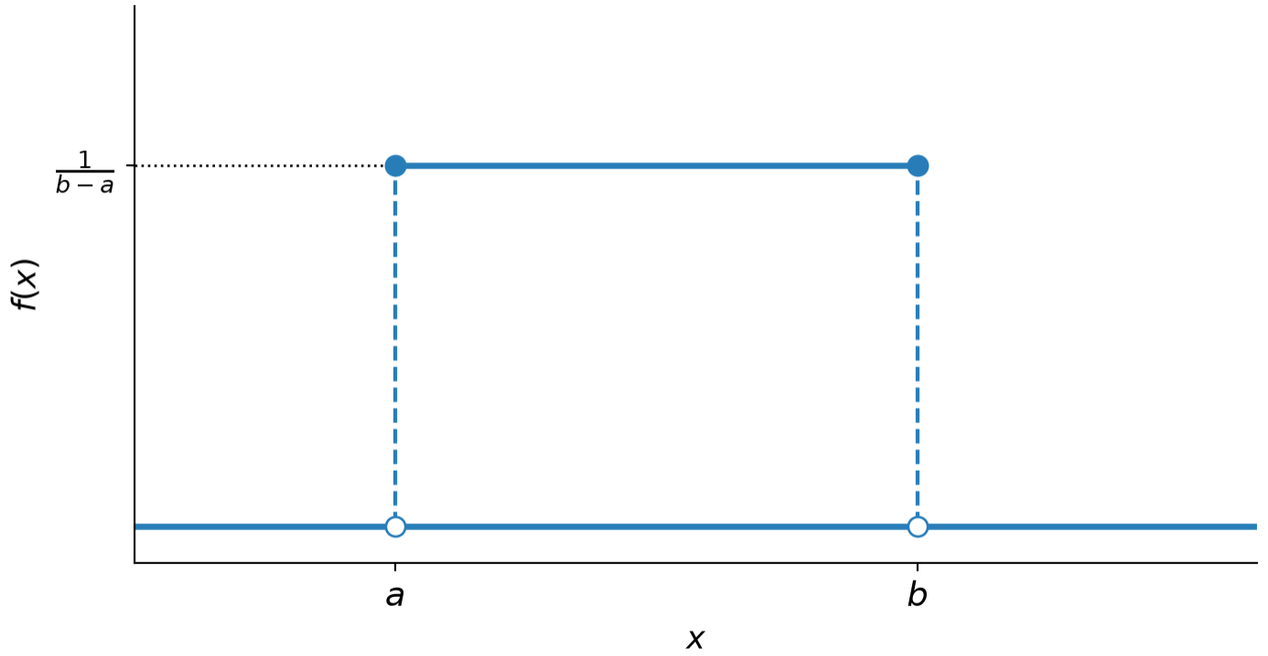
A continuous r.v. X is said to have the Uniform distribution on the interval (a, b) if its PDF is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

What processes follow the Uniform?

- Completely random number generation
- When every outcome is equally likely (note that $f(x)$ is the same for any value of x).

The PDF of the Uniform Distribution

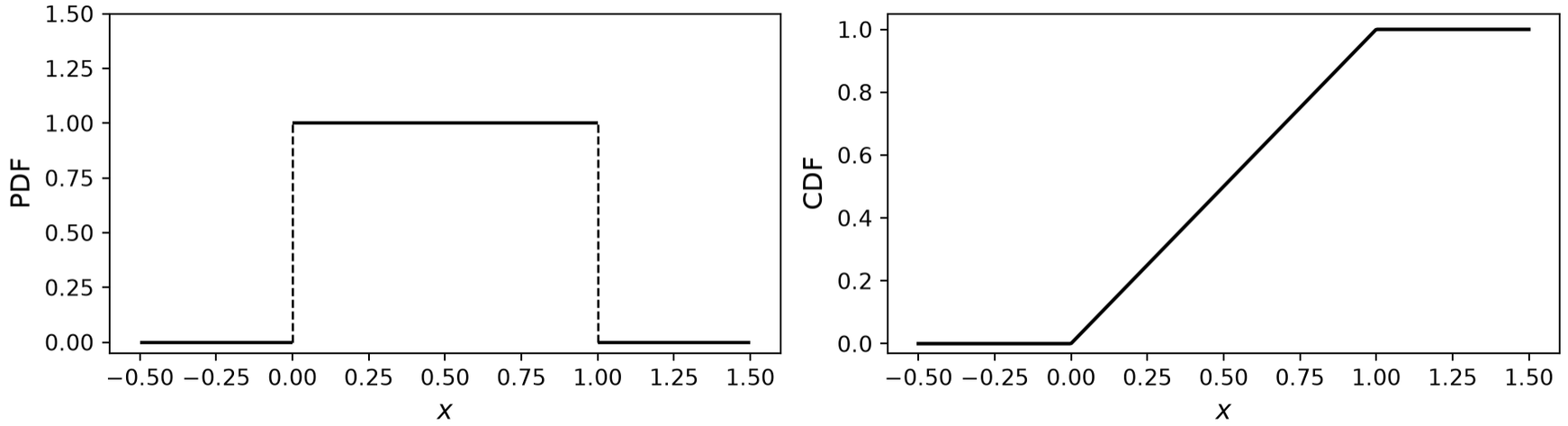


Quick exercise: Show that this is a valid PDF.

The CDF of the Uniform Distribution

The CDF of the Uniform is:

$$F(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x < b \\ 1 & \text{if } x \geq b \end{cases}$$



Let’s Find the Mean of the Uniform Distribution

Recalling that the mean is defined as:

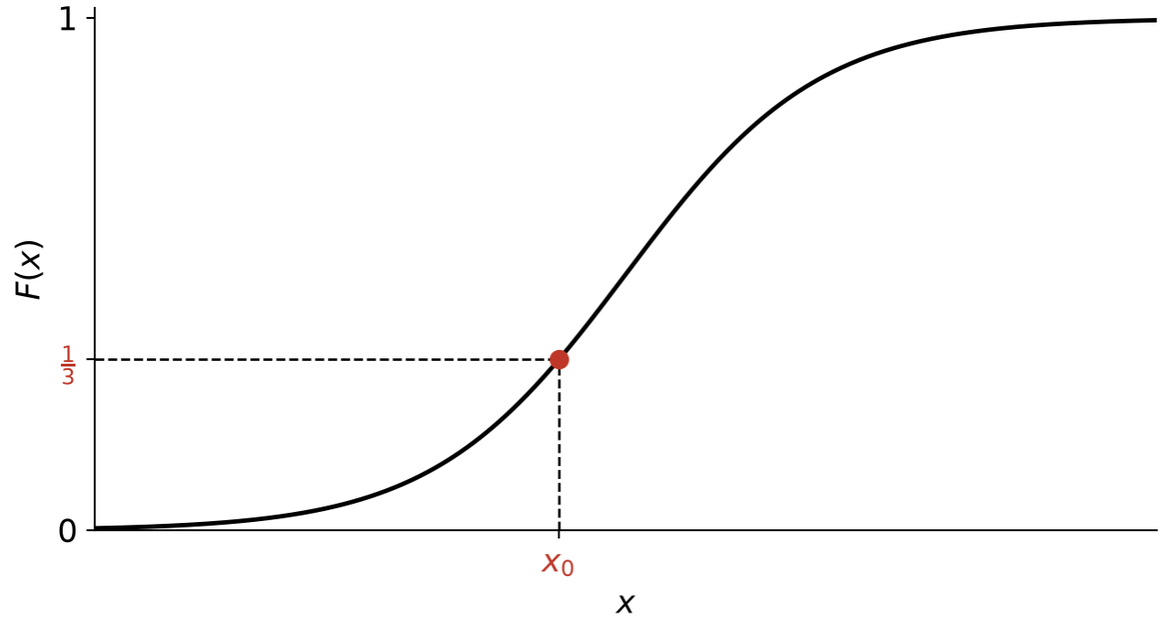
$$\int_{-\infty}^{\infty} x f(x)\, dx$$

We can plug in $f(x)$ for the Uniform (and focus on the support of X):

$$\begin{aligned}\int_a^b x\left(\frac{1}{b-a}\right) dx &= \left.\frac{x^2}{2(b-a)}\right|_a^b \\ &= \frac{b^2-a^2}{2(b-a)} \\ &= \frac{(b+a)(b-a)}{2(b-a)} = \frac{a+b}{2}\end{aligned}$$

Universality of the Uniform

Take a continuous X with CDF F, and let x_0 be the point where $F(x_0) = 1/3$:

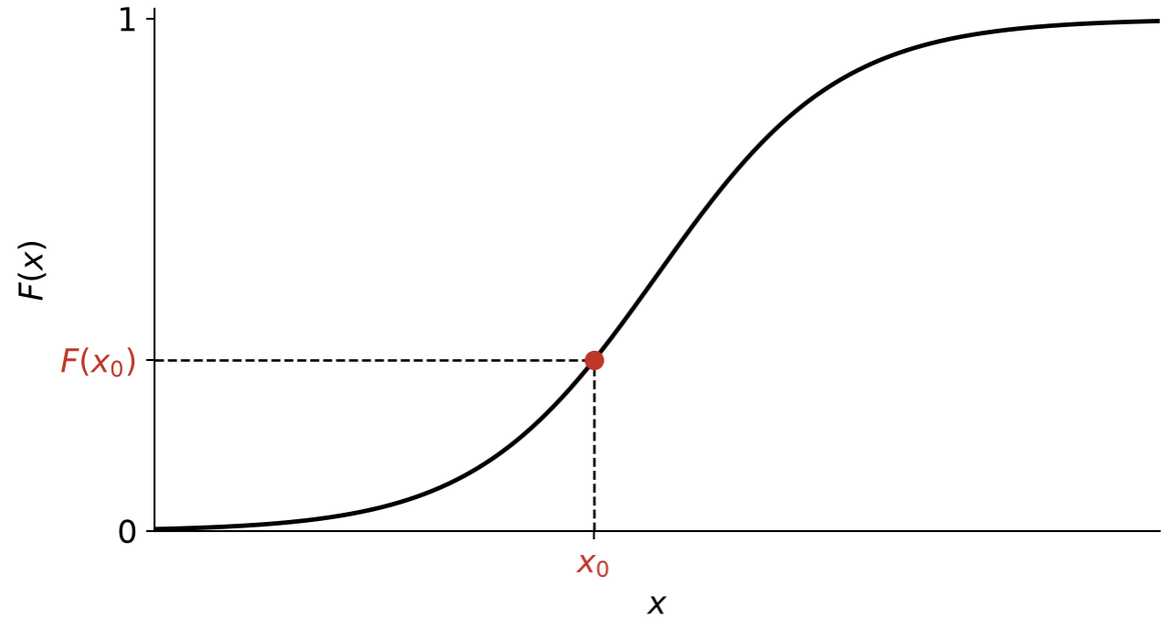


Now feed X into its *own* CDF and ask how often $F(X)$ lands below $1/3$:

$$P\big(F(X) \le 1/3\big) = P(X \le x_0) = F(x_0) = 1/3$$

The middle step is just the picture: F is increasing, so $F(X) \le 1/3$ says exactly the same thing as $X \le x_0$.

Universality of the Uniform



Nothing was special about $1/3$ - the same argument works at every level p:

$$X \sim F \quad \Longrightarrow \quad F(X) \sim \text{Unif}(0,1) \quad \Longrightarrow \quad X \sim F^{-1}(\text{Unif}(0,1))$$

Any continuous distribution, pushed through its own CDF, comes out Uniform on $(0,1)$.

Also known as probability integral transform, inverse transform sampling, the quantile transformation, and even the fundamental theorem of simulation.

Meet the Normal Distribution

The PDF of a Normal-distributed random variable is:

$$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

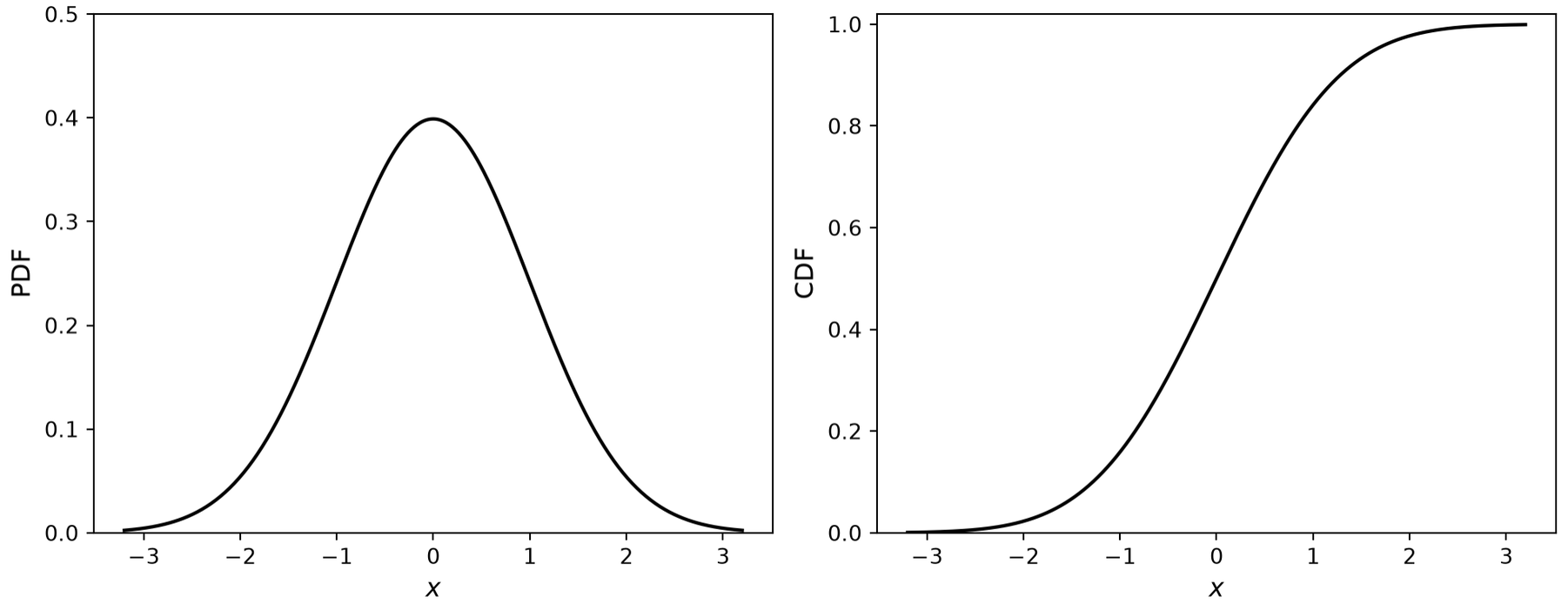
Notice that we have two parameters: μ (mean) and σ^2 (variance).

A special case of the Normal is the Standard Normal, which is just the Normal with $\mu = 0$ and $\sigma^2 = 1$:

$$\phi(x) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}x^2}$$

And by convention, the CDF of X is often written $\Phi(x)$. (The functional form of the CDF is ugly and you don't need to know it.)

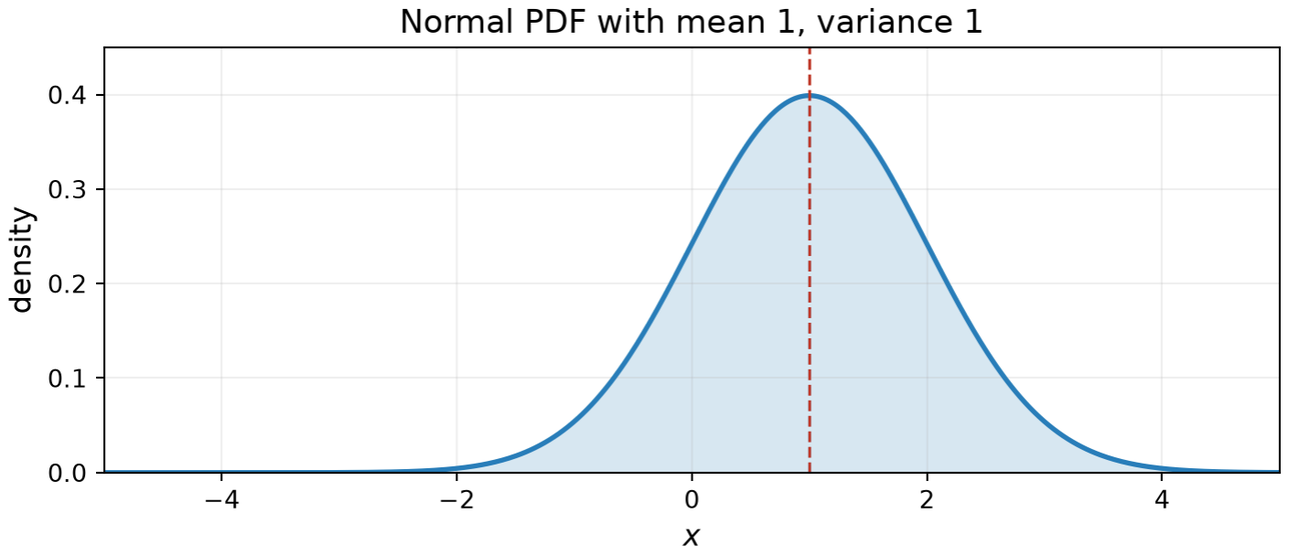
Meet the Normal Distribution



Standard Normal PDF ϕ (left) and CDF Φ (right).

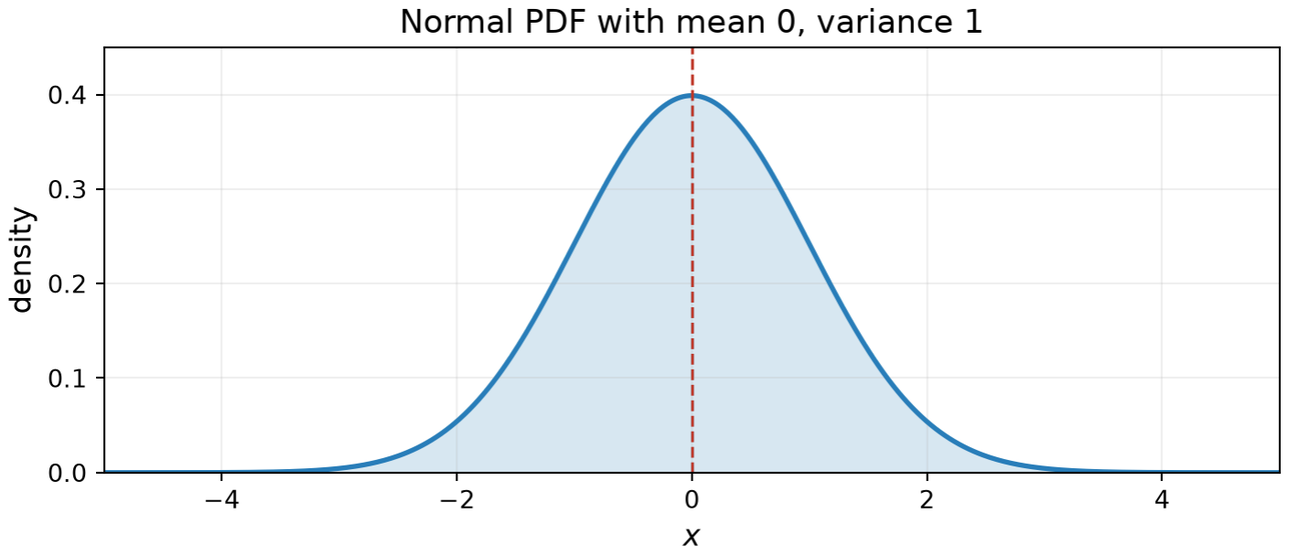
Parameters of the Normal Distribution

The mean μ is also called the location parameter: where the distribution is centered.



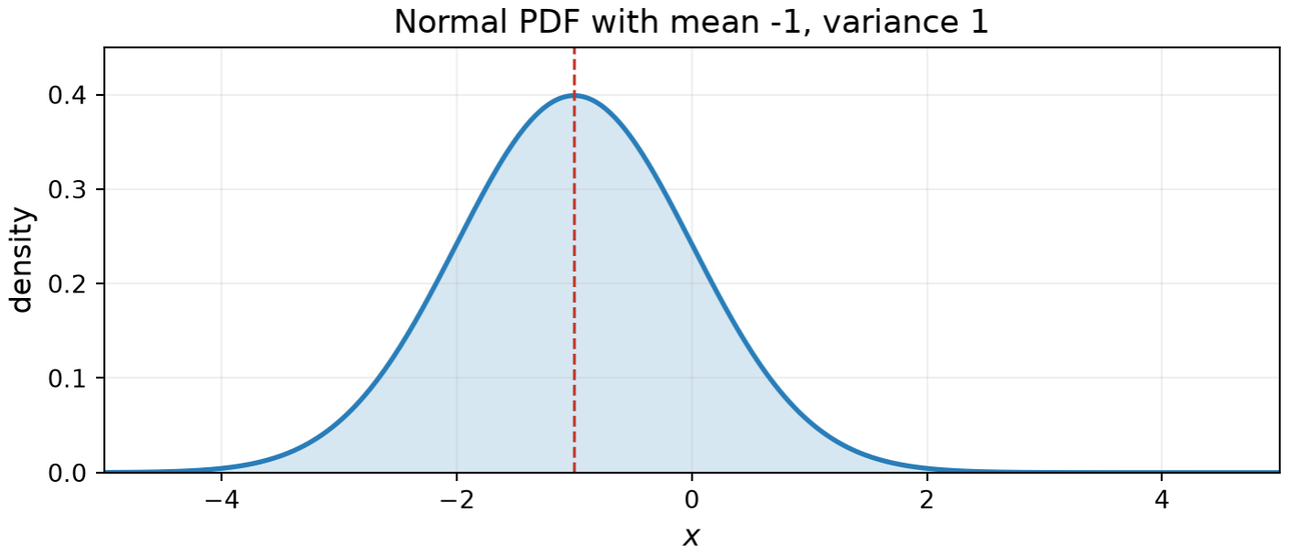
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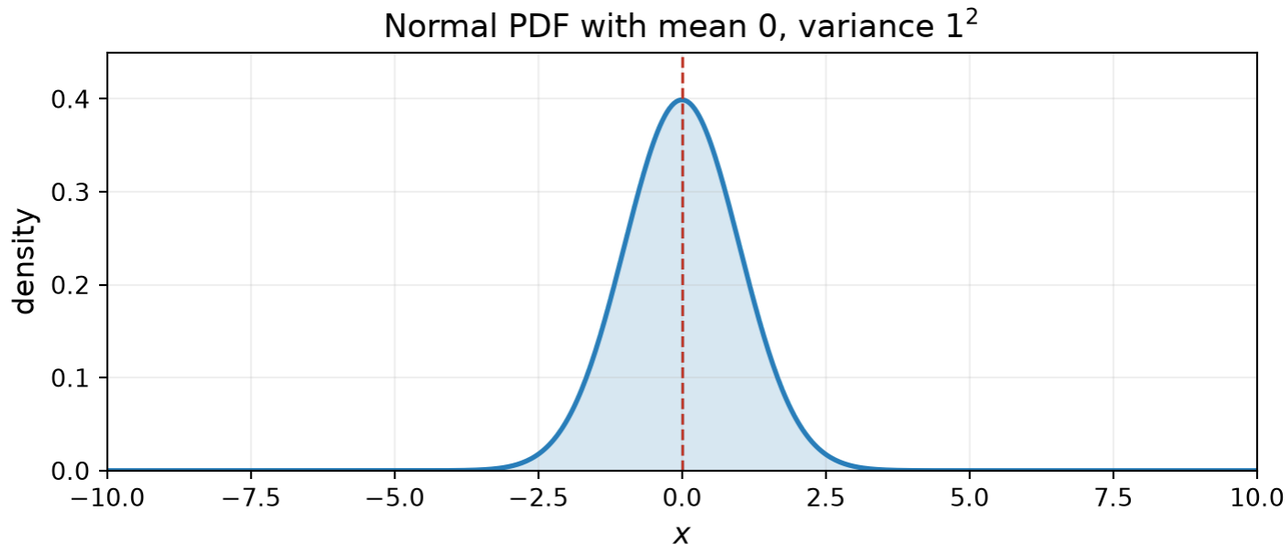
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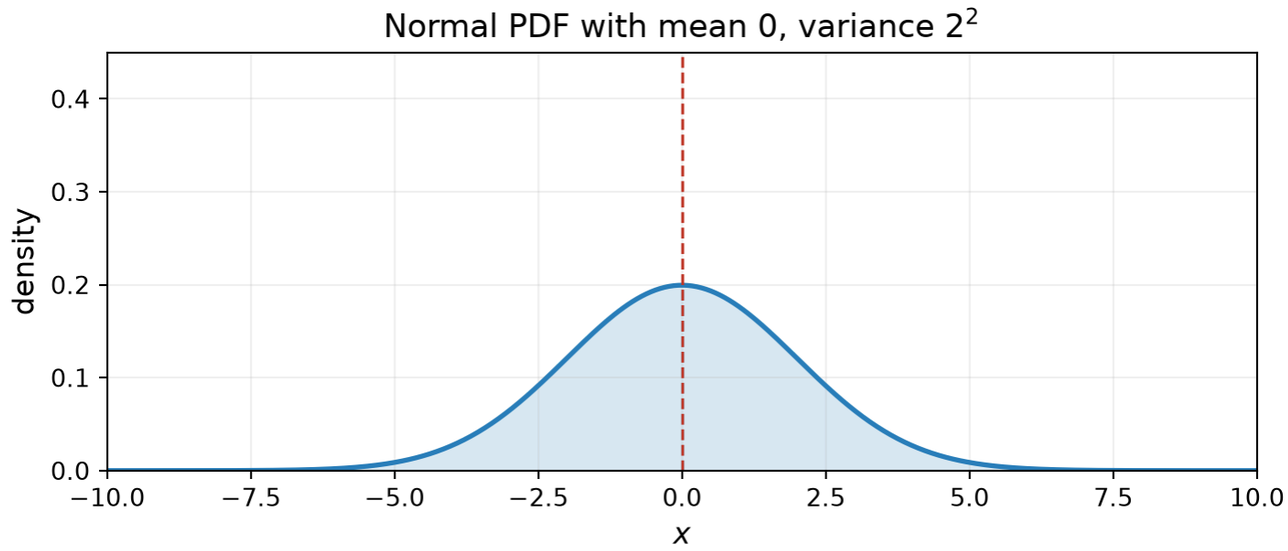
Parameters of the Normal Distribution

And the variance σ^2 is also called the scale parameter, which determines its shape.



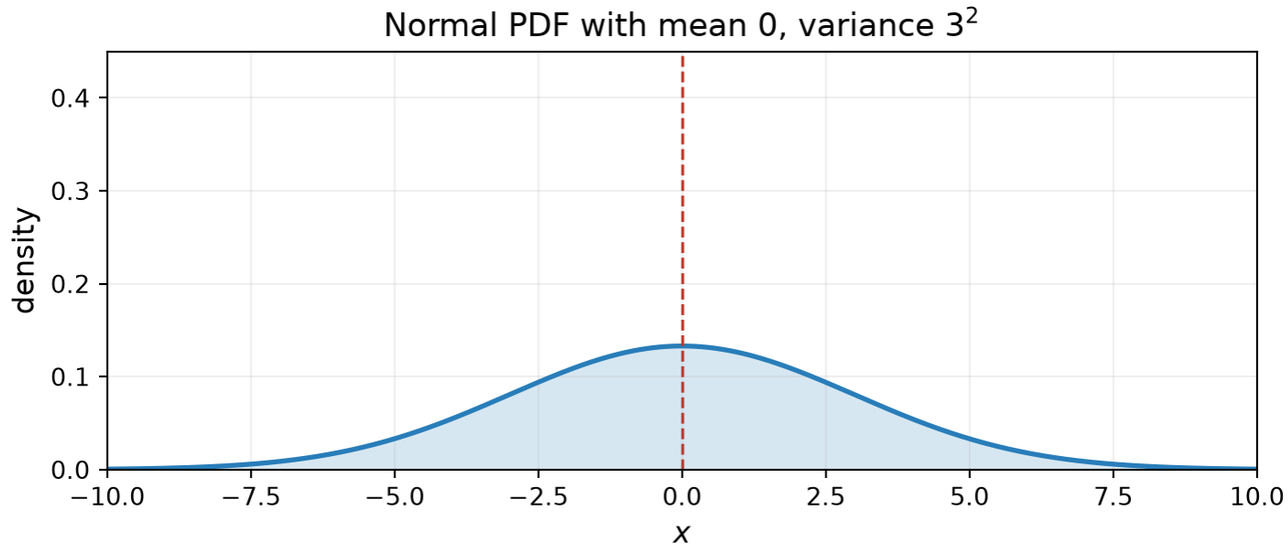
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Standardization

Any Normal-distributed random variable X with mean μ and variance σ^2 may be transformed into a Standard Normal:

$$\text{If } Z \sim \mathcal{N}(0,1), \text{ then } X = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$$

That means we can transform any Normal r.v. into a Standard Normal:

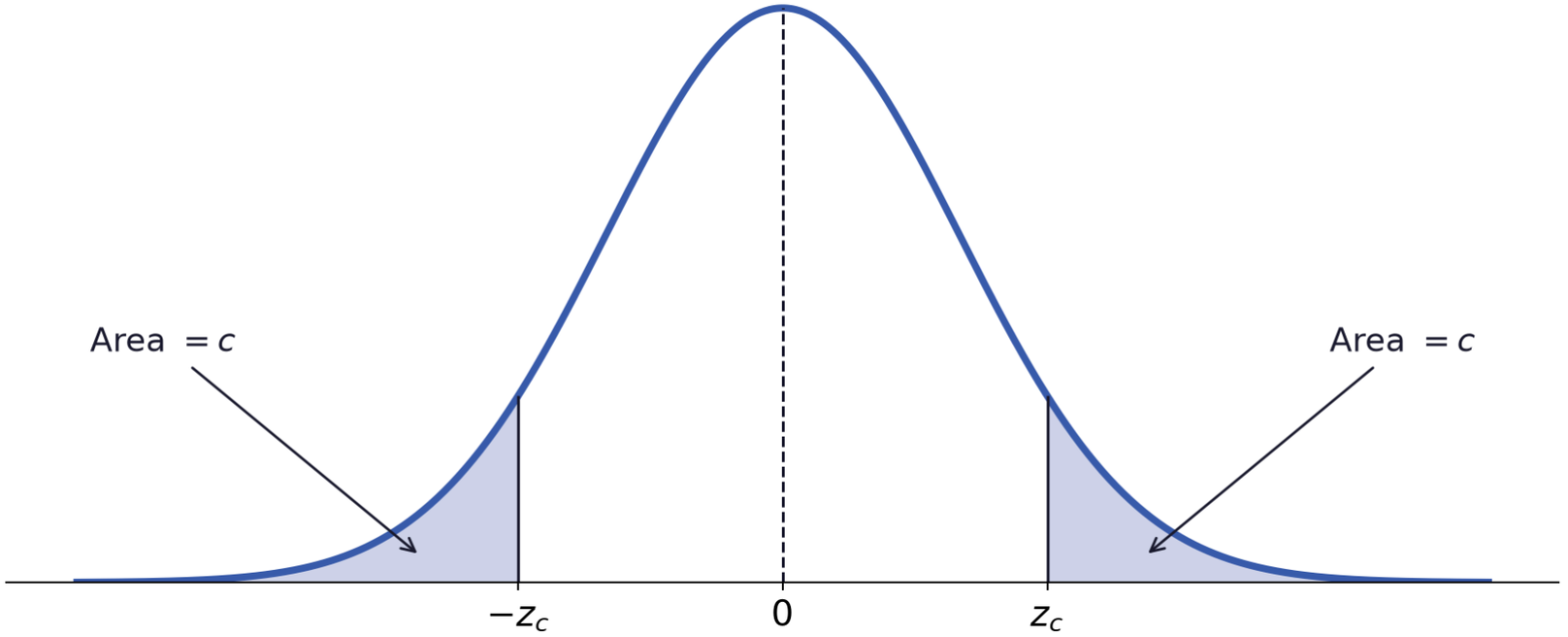
$$Z = \frac{X - \mu}{\sigma}$$

We often do this in our data to put our variables on the same scale.

Cool Stuff About the Normal

1. **The Central Limit Theorem!!!!**
    - *For independent and identically distributed random variables, the sampling distribution of the standardized sample mean tends towards the standard normal distribution even if the original variables themselves are not normally distributed.*
2. Symmetry of the PDF: $\phi(z) = \phi(-z)$
3. Symmetry of the tail areas: $\Phi(-z) = 1 - \Phi(z)$

Symmetry of the Normal PDF



68-95-99-rule

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then:

$$
\begin{aligned}
P(|X - \mu| < \sigma) &\approx 0.68 \\
P(|X - \mu| < 2\sigma) &\approx 0.95 \\
P(|X - \mu| < 3\sigma) &\approx 0.997
\end{aligned}
$$

Finally (for now), Meet the Exponential

The PDF of an Exponential-distributed random variable X is given by:

$$f(x) = \lambda e^{-\lambda x},\ x > 0$$

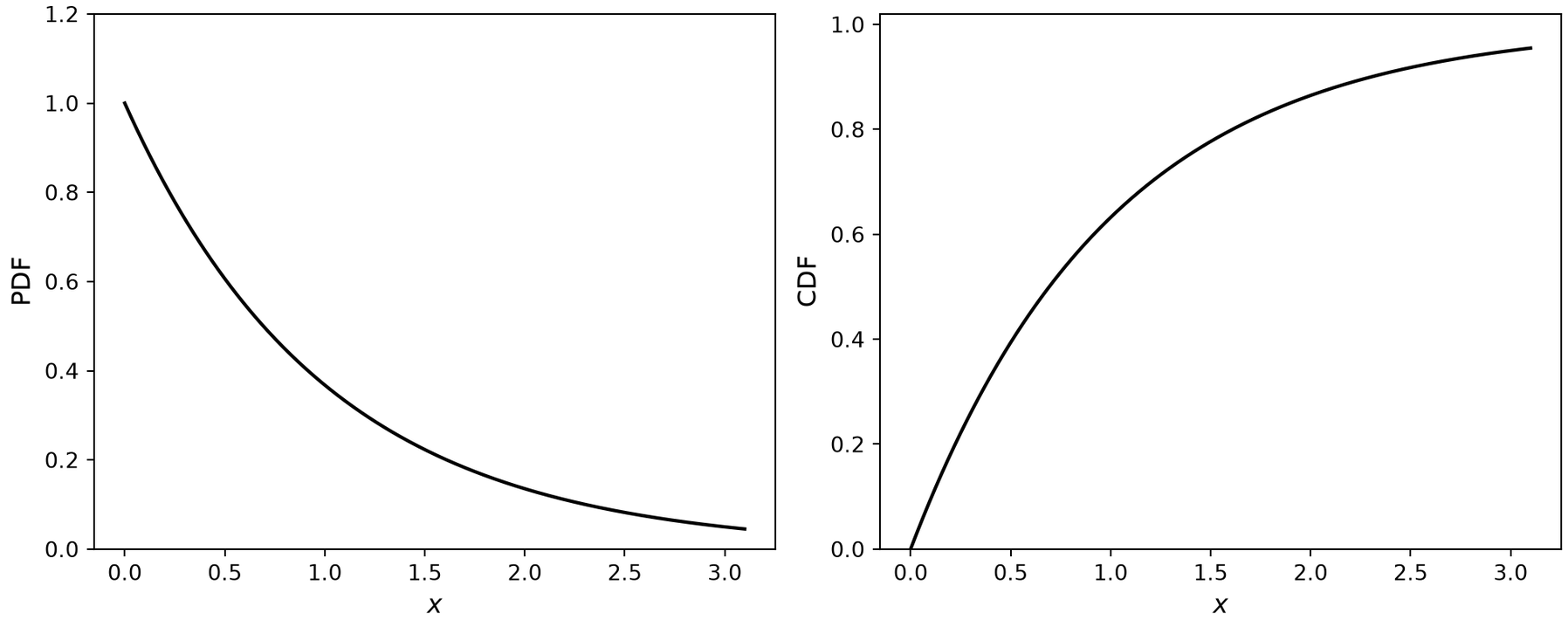
The only parameter here is λ, so we write $X \sim \text{Expo}(\lambda)$.

The CDF of X is given by:

$$F(x) = 1 - e^{-\lambda x},\ x > 0$$

(We can quickly differentiate this and check.)

Meet the Exponential



Expo(1) PDF and CDF.

Meet the Exponential

The Exponential is the continuous analog of the Geometric.

- Recall that a Geometric r.v. counts the number of failures until the first success in a sequence of Bernoulli trials
- An Exponential r.v. represents the continuous time you have to wait before the arrival of the first success
- Then the parameter λ can be interpreted as the rate of successes per some unit of time.
- Note also the connection to the Poisson (also part of the Exponential family), which is the discrete number of events per some unit of time

Cool Stuff About the Exponential

The Exponential displays the memoryless property:

$$P(X \geq s + t \mid X \geq s) = P(X \geq t)$$

The time you've already spent waiting has no effect on the time you will spend waiting for the event.

In other words, after you've already waited s minutes, the probability you'll have to wait another t minutes is the same as the probability of having to wait t minutes with no previous waiting time under your belt.

Implications of Memorylessness

What are the implications of memorylessness?

- If human lifetimes were Exponential, your life expectancy at age 80 would be the same as if you were a newborn baby.
- If the lifetime of a machine was Exponential, its time to failure would be the same no matter how long it's been operational (no wear and tear)

What behaves this way?

- Radioactive decay
- …?

But more importantly, the Exponential is just a building block for more flexible distributions that do account for the passage of time (e.g. the Weibull).

Part III: Continuous Joint Distributions

Moving On…

Let’s cover the continuous versions of:

1. Joint, Marginal, and Conditional distributions
2. Bayes’ Rule and the Law of Total Probability
3. Independence of continuous r.v.s
4. The expectation of a random variable that is a function of two continuous random variables (LOTUS)
5. Covariance and correlation

Joint Distribution of Continuous Random Variables

If X and Y have a continuous joint distribution, we require that the joint CDF:

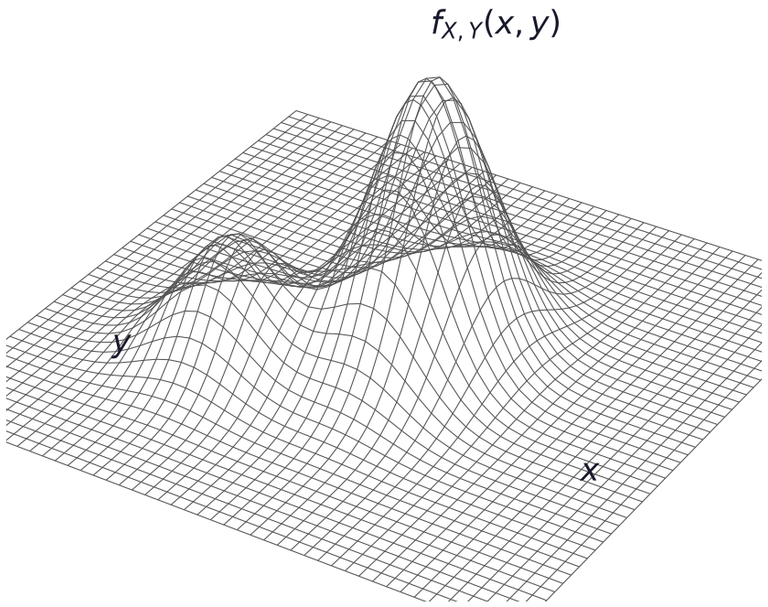
$$F_{X,Y}(x, y) = P(X \leq x,\, Y \leq y)$$

be differentiable with respect to both x and y. The partial derivative with respect to x and y is the joint PDF:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x, y)$$

(Let's try taking the partial derivatives of $F(x, y) = \frac{1}{2}x^2y^3$.)

Joint Distribution of Continuous Random Variables



Joint PDF of continuous r.v.s X and Y.

Marginal Distribution of Continuous Random Variables

To get the marginal distribution of X, integrate over all values of Y rather than summing as we did for discrete r.v.s:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$$

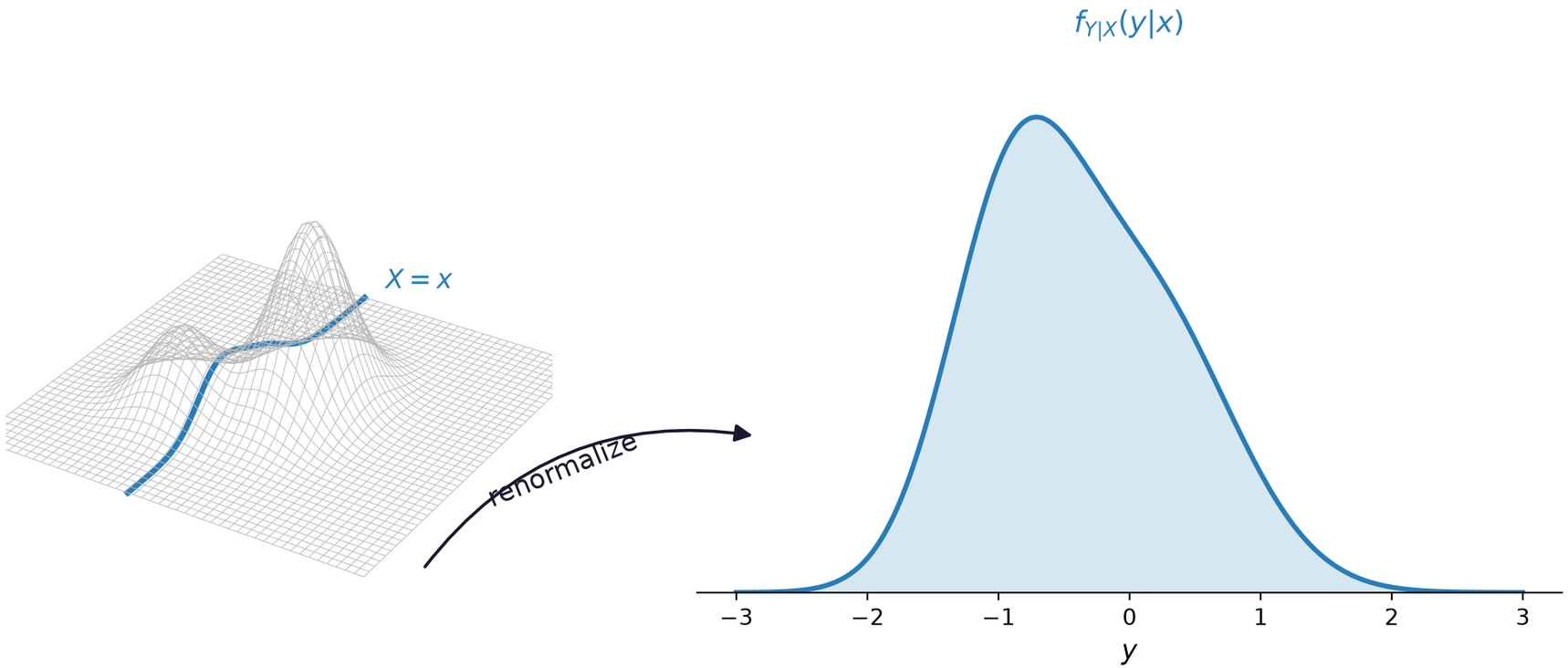
Conditional PDF

For continuous r.v.s with joint PDF $f_{X,Y}$, the conditional PDF of Y given $X = x$ is:

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

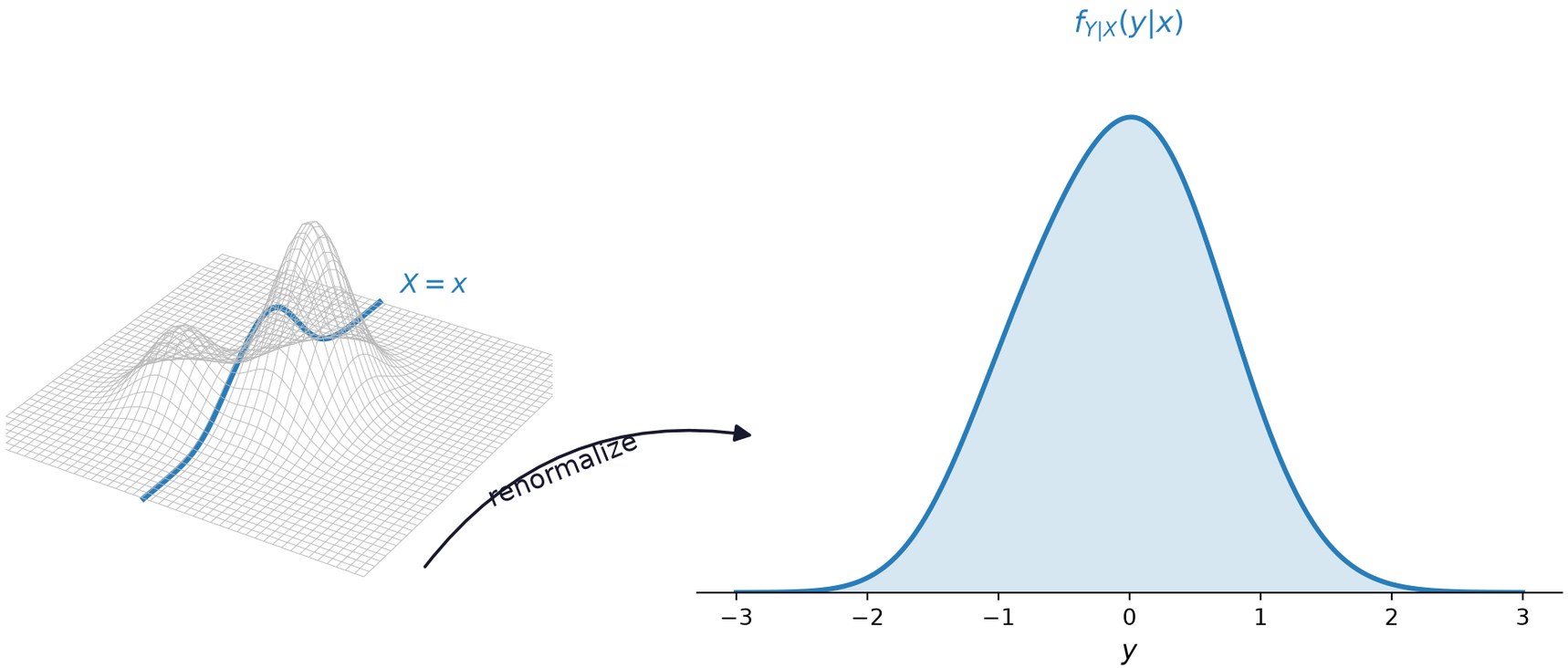
As with discrete random variables, the conditional is the joint over the marginal.

Conditional PDF



The conditional PDF is obtained by renormalizing the slice of the joint PDF at the fixed value x.

Conditional PDF



The conditional PDF is obtained by renormalizing the slice of the joint PDF at the fixed value x.

Note

How can we condition on $X = x$ for a continuous r.v. when we learned that this event has probability 0?

- Technically, we're conditioning on the event that X falls in some small interval containing x, say $\left(x - \varepsilon,\ x + \varepsilon\right)$, and then taking the limit as ε goes to 0.
- (Glad you asked!)

Bayes' Rule and the LOTP Carry Over Neatly

For continuous r.v.s X and Y, we have Bayes' Rule as:

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y)\, f_Y(y)}{f_X(x)}$$

And the Law of Total Probability as:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x \mid y)\, f_Y(y)\, dy$$

(We have simply replaced the sum from discrete world with an integral.)

You Can Combine Discrete and Continuous Random Variables

No problem! Here are the four versions of Bayes' rule, summarized in a table.

	Y **discrete**	Y **continuous**
X **discrete**	$P(Y=y \mid X=x) = \frac{P(X=x\|Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y \mid X=x) = \frac{P(X=x\|Y=y)f_Y(y)}{P(X=x)}$
X **continuous**	$P(Y=y \mid x) = \frac{f_X(x\|Y=y)P(Y=y)}{f_X(x)}$	$f_{Y\|X}(y \mid x) = \frac{f_{X\|Y}(x\|y)f_Y(y)}{f_X(x)}$

Part IV: Covariance and Correlation

Covariance

The covariance between two random variables X and Y is defined as:

$$\operatorname{Cov}(X, Y) = E\big((X - EX)(Y - EY)\big)$$

By the linearity of expectation, this can be written as:

$$\operatorname{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

Think of covariance as a measure of whether the two variables move together.

Some Covariance Rules

1. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
2. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
3. $\mathrm{Cov}(X, c) = 0$ for any constant c
4. $\mathrm{Cov}(aX, Y) = a\,\mathrm{Cov}(X, Y)$ for any constant a
5. $\mathrm{Cov}(X + Y, Z) = \mathrm{Cov}(X, Z) + \mathrm{Cov}(Y, Z)$

   $\Rightarrow \mathrm{Cov}\left(\sum_{i=1}^{m} a_i X_i, \sum_{j=1}^{n} b_j Y_j\right) = \sum_{i,j} a_i b_j \mathrm{Cov}(X_i, Y_j)$
6. $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$

   $\Rightarrow \mathrm{Var}\left(\sum_{i=1} X_i\right) = \sum_i \mathrm{Var}(X_i) + 2\sum_{j<i} \mathrm{Cov}(X_i, X_j)$

Correlation

The correlation between two random variables is given by:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

This scaling puts correlation between -1 and 1 no matter the scale of X and Y, making correlation easy to interpret.

Correlation Imposes Linearity

An important note: if X and Y are independent, then they are uncorrelated.

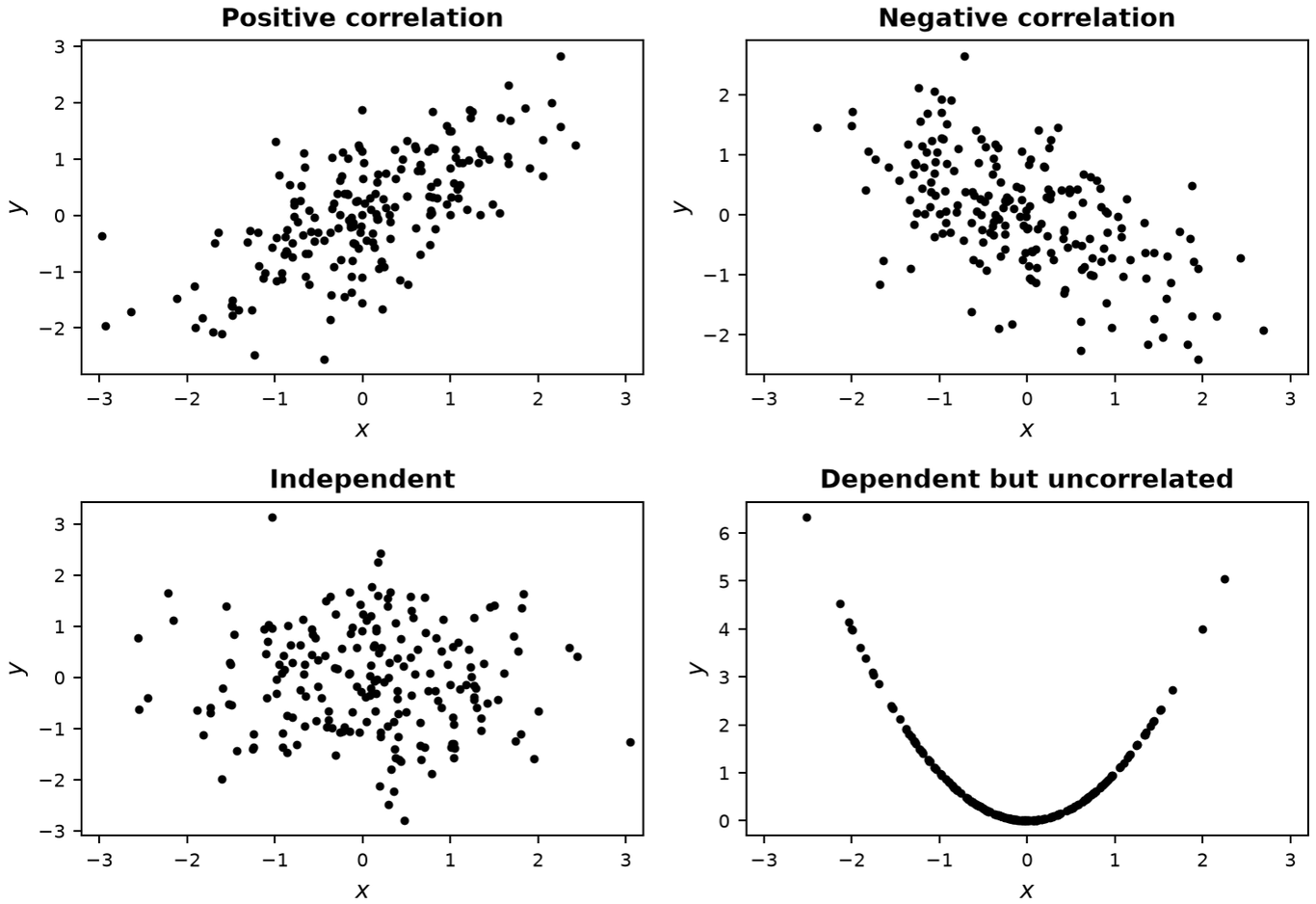
Proof: $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$. For independent r.v.s, $E(XY) = E(X)E(Y)$. Thus, $\mathrm{Cov}(X,Y)$ is 0 and so is correlation.

However, if two variables are uncorrelated, then they might not be independent. Consider:

$$X \sim \mathcal{N}(0,1)$$
$$Y = X^2$$

These are perfectly dependent. But, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0 - 0$.

Correlation Imposes Linearity



Summary - continuous random variables

Concept	Key formula
PDF	$f(x) = F'(x)$, with $f \geq 0$ and $\int f = 1$
Interval probability	$P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$
Expectation	$E(X) = \int_{-\infty}^{\infty} xf(x)dx$
Uniform(a, b)	$f(x) = \frac{1}{b-a}$ on (a, b); $E(X) = \frac{a+b}{2}$
Normal	$\phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$; $Z = \frac{X-\mu}{\sigma}$
Exponential	$f(x) = \lambda e^{-\lambda x}$; memoryless
Joint / marginal PDF	$f_{X,Y} = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}$; $f_X(x) = \int f_{X,Y}(x, y)dy$
Conditional PDF	$f_{Y\|X}(y \mid x) = f_{X,Y}(x, y)/f_X(x)$
Covariance	$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$
Correlation	$\mathrm{Corr}(X, Y) = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$

Further Reading

Blitzstein & Hwang, *Introduction to Probability*

- Ch. 5.1–5.2 (p. 213–223) - PDFs, CDFs, the Uniform
- Ch. 5.3 (p. 224–228) - Universality of Uniform
- Ch. 5.4–5.5 (p. 231–244) - the Normal and the Exponential
- Ch. 7 p. 312–320 (stop before Example 7.1.25) - continuous joint distributions
- Ch. 7.3 (p. 326–332) - covariance and correlation

Further Self Practice

Blitzstein & Hwang, *Introduction to Probability*

- CDF, PDF for Hybrid distributions: Example 7.1.26
- 2D LOTUS: Examples 7.2.2 and 7.2.3
- Strategic Practice 6.1
- Strategic Practice 7
- Strategic Practice 8.1

Solutions to the practice problems at projects.iq.harvard.edu/stat110